Senior Associate, Risk and Structuring Analytics

Clearway Energy Group · Houston, Texas, United States

Job ID (this archive)	ENERGY-105
Employer	Clearway Energy Group
Location(s)	Houston, Texas, United States
Career stage (archive tag)	Experienced (Mid-career)
Location group	Texas (outside DFW)
Majors (archive tags)	Data Science
Energy area	Renewable energy / analytics
Pay listed	Not listed
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://job-boards.greenhouse.io/clearwayjobs/jobs/5250917007
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-076

Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

What The Role Is

Clearway Energy seeks a Senior Associate of Risk and Structuring Analytics to develop top-notch quantitative models that drive BESS valuation, structured products pricing, gas generation optimization, and power market analysis, while also quantifying and monitoring the risk exposures those positions create. In this role, you will build and enhance Clearway's internal quantitative modeling platform (CWENQuant), develop advanced BESS dispatch optimization and price forecasting models, and provide independent risk analytics — mark-to-market, value-at-risk, credit, and scenario/stress analysis — that directly inform commercial decisions across trading, origination, development, and asset management.

What You'll Be Doing

Quantitative/Economic Modeling

- Develop and deploy to production complex optimization models and quantitative analysis for BESS dispatch optimization, renewables, gas generation, and data center infrastructure projects across the US.
- Develop Stochastic or Machine Learning / Deep Learning models to forecast DA, Ancillary, and RT prices based on fundamental variables such as load, gas prices, and wind + solar generation to inform trading.
- Develop and maintain gas dispatch, HRCO, toll, and other structured pricing models; perform mark-to-market analysis; support due diligence, underwriting, and credit analytics.
- Refine vendor models or develop in-house models for capacity expansion with a view on market policy (RPS/GHG reduction), LCOEs, retirements, transmission, reliability, and demand growth.
- Benchmark models against actual operations and market outcomes to improve forecast accuracy and optimization.
- Build and enhance CWENQuant; develop data pipelines with internal and external resources; create automated workflows and market data dashboards using data science and optimization techniques.
- Design and maintain mark-to-market (MtM), Value-at-Risk (VaR), and Profit-at-Risk (PaR) frameworks across BESS, gas, and structured product positions, independent of the deal/trading desks that originate them.
- Run scenario, sensitivity, and stress-testing analyses (price shocks, volatility regimes, basis blowouts, extreme weather/renewable generation outcomes) to quantify tail risk in the portfolio.
- Quantify hedge effectiveness and basis risk for structured transactions, PPAs, and tolling/HRCO arrangements; recommend hedge adjustments.

What You'll Bring

- Bachelor’s degree in Engineering, Finance, Mathematics, Statistics, Economics, or a related field.
- 3+ years of quantitative or modeling experience within the energy sector, including trading firms, banks, utilities, IPPs, ISOs, or BESS companies.
- Working experience developing ML/DL models using scikit-learn, Keras, or TensorFlow.
- Experience with linear/Mixed Integer programming and associated tools like PuLP, Gurobi, CPLEX, etc.
- Strong quantitative background with deep hands-on experience coding in Python.
- Knowledge of BESS (strongly preferred), renewable, and conventional (gas) assets, HRCO, and toll modeling.
- Strong understanding of power market economics, wholesale market structure, and asset operations in ISO (CAISO, ERCOT, etc.) markets.
- Proven ability to build top-notch optimization models and stochastic or ML/DL forecasting frameworks such as price, load, or renewable generation forecasting.
- Understanding of option pricing, volatility, game theory, opportunity cost, and optimization.
- Ability to formulate and articulate viewpoints (written and verbal) in a clear, persuasive, and succinct manner.
- Ability to handle multiple concurrent efforts and provide high-quality deliverables accurately.
- Ability to work well in a fast-paced, team-oriented, collaborative environment that emphasizes attention to detail, meeting deadlines, and working together to achieve company-wide objectives.
- Strong interpersonal, analytical, and problem-solving skills.

What Would Be Nice

- Programming skills required: Python is a must; R, SQL, etc. are a plus.
- Working experience with stochastic models, mean reversion, and jump diffusion preferred.
- Exposure to ETRM systems, MtM, and VaR model validation/governance practices is a plus.
- Master’s or PhD is preferred.

No visa sponsorship or any kind of sponsorship for employment authorization is available for this role, now or in the future.

#LI-Hybrid

The pay rate for the successful candidate will depend on geographic location, skills, relevant and demonstrated experience, education, training and certifications, and other factors permitted by law. This role is eligible to earn an annual cash bonus, subject to personal and company performance goals.

Salary Range Across all U.S. Locations

$140,000—$178,000 USD

Clearway Energy Group is leading the transition to a world powered by clean energy. Along with

our public affiliate Clearway Energy, Inc., our portfolio comprises approximately 11.6 GW of gross generating capacity in 26 states, including 9.1 GW of wind, solar, and battery energy storage assets, and over 2.5 GW of conventional dispatchable power generation providing critical grid reliability services. As we develop a nationwide pipeline of new energy projects for the future, Clearway’s operating fleet generates enough reliable electricity to power more than 3 million homes today. Clearway Energy Group is headquartered in San Francisco with offices in Denver, Houston, Phoenix, Princeton, and San Diego. For more information, visit clearwayenergygroup.com.

Our Commitment to Diversity, Equity, & Inclusion

Clearway Energy Group’s vision is a world powered by clean energy. Along with our public affiliate Clearway Energy, Inc., our portfolio comprises approximately 11.8 GW of gross generating capacity in 26 states, including 9.1 GW of wind, solar, and battery energy storage assets, and over 2.8 GW of flexible dispatchable power generation providing critical grid reliability services. As we develop a nationwide pipeline of new energy projects for the future, Clearway’s operating fleet generates enough reliable electricity to power more than 2 million homes today. Clearway Energy Group is headquartered in San Francisco with offices in Denver, Houston, Phoenix, Princeton, and San Diego. For more information, visit clearwayenergygroup.com.

Working at Clearway

Here at Clearway, we’re committed to balancing flexibility while fostering strong relationships with our teammates. We do this by prioritizing corporate new hires based near one of our offices in San Francisco, San Diego, Scottsdale, Houston, Denver, & Princeton. Our office-based employees work together from our offices on Tuesday, Wednesday & Thursday to collaborate &

learn, build community, get to know one another, & enjoy company-provided meals & events. Our site-based employees collaborate in-person daily to keep our locations operating safely and efficiently while fostering strong relationships within their local communities.

Clearway Energy is an equal opportunity employer that values a broad diversity of talent, knowledge, experience & expertise. We intentionally foster a culture of inclusion that empowers our employees to deliver superior performance to the communities we serve. We encourage minorities, women, individuals with disabilities & protected veterans to join the team. Clearway is a proud promoter of employment opportunities for our Military & Veterans.

What We Provide

Clearway offers all eligible employees working 20+ hours per week a comprehensive menu of benefits: generous PTO, medical, dental & vision care, HSAs with company contributions, health FSAs, dependent daycare FSAs, commuter benefits, relocation, & a 401(k) plan with employer match, a variety of life & accident insurances, fertility programs, adoption assistance, generous parental leave, tuition reimbursement, & benefits for employees in same-sex marriages, civil unions & domestic partnerships. For more on Clearway benefits, visit our Benefits Website.

Notice to Applicants

Clearway Energy is strongly committed to a philosophy of equal opportunity and fully complies with equal employment laws, ordinances, and regulations for all applicants and employees. We will make reasonable accommodations to enable individuals with disabilities to apply and compete for employment opportunities for which they are qualified. If you need a reasonable accommodation because of a disability for any part of the employment process, please send an e-mail to recruiting@clearwayenergy.com and let us know the nature of your request and your contact information.

All official communication regarding this position will come only from an email address ending with @clearwayenergy.com. If you receive outreach from any other email address claiming to represent Clearway Energy, please do not respond, as it is fraudulent.

Notice to California Applicants Clearway Energy is strongly committed to a philosophy of equal opportunity and fully complies with equal employment laws, ordinances and regulations for all applicants and employees.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: Bachelor’s degree in Engineering, Finance, Mathematics, Statistics, Economics, or a related field. | 3+ years of quantitative or modeling experience within the energy sector, including trading firms, banks, utilities, IPPs, ISOs, or BESS companies. | Master’s or PhD is preferred.

Key skills listed in the archive: Python; SQL; R; TensorFlow; machine learning; statistics; forecasting; optimization

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer Greenhouse Employer / ATS identifier: 5250917007

Career stage: Mid-career

Major fit: Direct skills pathway: data science, machine learning, analytics or data engineering; exact degree/experience requirements retained. Energy-sector connection: Renewable-energy analytics, risk models and power-market analysis.

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer Greenhouse. Source identifier: 5250917007.

Sources and retrieval details

Employer posting

https://job-boards.greenhouse.io/clearwayjobs/jobs/5250917007

Retrieved source

https://boards-api.greenhouse.io/v1/boards/clearwayjobs/jobs?content=true

Retrieved: 2026-10-26T16:34:56.524900+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.